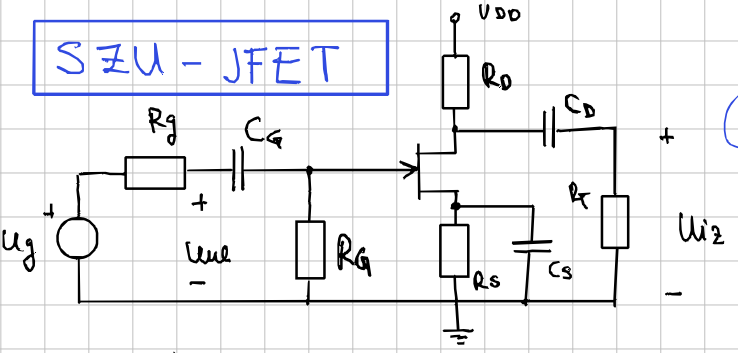
NISKO FREKVENCIJSKA analiza

SZU - JFET



→ od srednjih do niskih f.

1 ↳ C_G, C_D = serijski

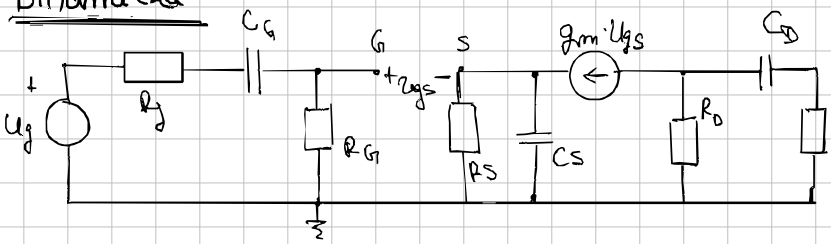
$\omega \downarrow\downarrow \sim Z_C = \infty \left(\frac{j\frac{\omega}{\omega_1}}{1+j\frac{\omega}{\omega_1}} \right)$

2 ↳ C_S = premosni

$\omega \uparrow\uparrow \sim Z_C = 0$

$\omega \downarrow\downarrow \sim Z_C = \infty \left(\frac{1+j\frac{\omega}{\omega_1}}{1+j\frac{\omega}{\omega_2}} \right)$

Dinamička:



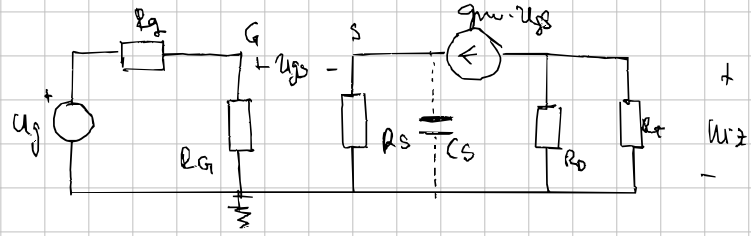
*odvojeno gledamo to dvoje

- U području niskim frekvencija do izražaja dolaze kapaciteti koji su u pojačalu spojeni serijski signalu
- Impedancija kapaciteta koji su spojeni paralelno signalu i koji guše prolaz signala na VF → u NF su vrlo velike → u NF = otvoreni krug

CS-utjecaj C_G i C_D predstavljaju K.S. → $A_V = A_{Vn} \cdot \frac{1+j\frac{\omega}{\omega_1}}{1+j\frac{\omega}{\omega_2}}$

↳ gledamo ga na najnižim frekvenc.

A_{Vn} = pojačanje na NF kada se još uvijek vidi R_S jer C_S još ne djeluje + C_D i C_G po pretpostavci K.S.



$Z_C = \frac{1}{j\omega C_S}$ → $\omega \to 0$: $\frac{1}{0} = \infty$ P.H.; $\omega \to \infty$: $\frac{1}{\infty} = 0$ K.S.

$\boxed{\omega \to 0}$ Odspojeni C_S → $\boxed{A_{Vn} = \dots = \frac{-g_m (R_D \| R_T)}{1+g_m \cdot R_S}}$ → degeneracije koja ruši pojačanje

$\boxed{\omega \to \infty}$ K.S. C_S → $\boxed{A_{V0} = -g_m (R_D \| R_T)}$ → na višim frekv. (srednjim) nema degeneracije

$$\boxed{A_{vn} = \cdots = \frac{-g_m (R_D \| R_T)}{1 + g_m \cdot R_S}}$$

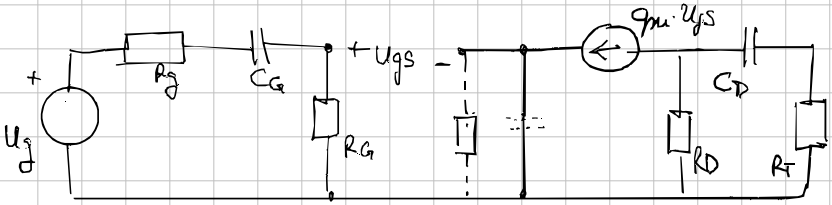
degeneracija koja ruši pojačanje

→ Što još ruši pojačanje?

SVAKI VEZNI C

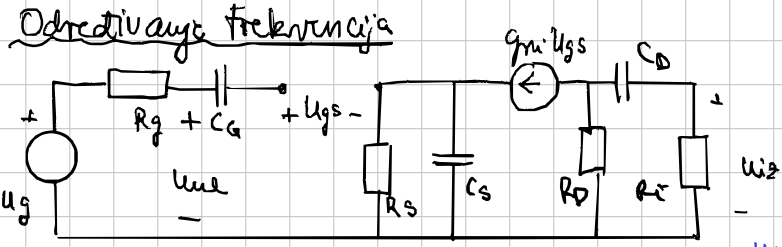
Utjecaj C_G i C_D

*sada promatramo za $C_S = K.S.$



Budući da C_G i C_D ruše pojačanje onda A_{vn} (jer dolaze dvojno.)

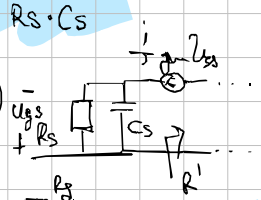
Određivanje frekvencija



$$A_v(j\omega) = A_{vn} \cdot \frac{1 + j\frac{\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_2}} \cdot \frac{j\frac{\omega}{\omega_3}}{1 + j\frac{\omega}{\omega_3}} \cdot \frac{j\frac{\omega}{\omega_4}}{1 + j\frac{\omega}{\omega_4}}$$

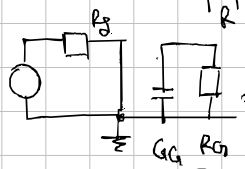
$$\omega_1 = \frac{1}{\tau_{s_1}} = \frac{1}{R_S \cdot C_S}$$

$\omega_2 = (C_G = \text{masa})$, $u_{ul} = 0$



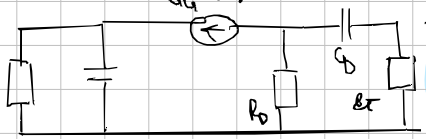
$$= \frac{1}{\tau_{s_2}} = \frac{1}{C_S \cdot \left(R_S \| \frac{1}{g_m}\right)}$$

$$\omega_3 = \frac{1}{\tau_G} =$$



$$= \frac{1}{C_G \cdot R_G}$$

$$\omega_4 = \frac{1}{\tau_D} =$$



$$\frac{1}{\tau_D} = \frac{1}{C_D (R_D + R_T)}$$

SULGOJ 2025

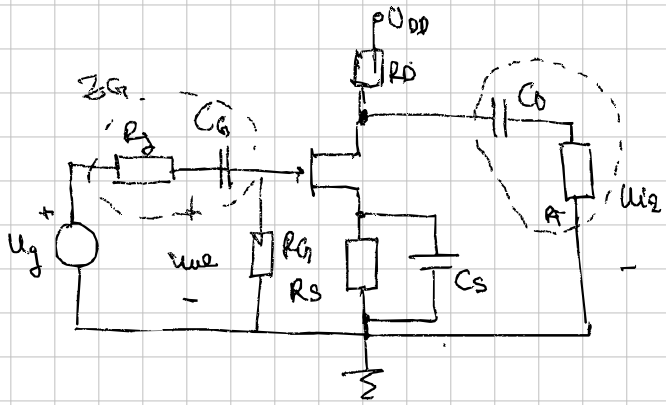
2. JFET NF i VF

→ na kojim frekv utječu C_G i C_D (vezni)

→ nacrtati amp. frek karakt

→ izraz za pojačanje

→ frekvencije



C_G i C_D su vezni kond

$\omega \uparrow\uparrow$; $Z_C = \frac{1}{j\omega C} = \frac{1}{\infty} = \emptyset$ K

↳ K.S. na višim f.

$\omega \downarrow\downarrow$; $Z_C = \frac{1}{\emptyset} = \infty$ P.H.

odspojeni na najnižim

Vezni kond: $\frac{j\frac{\omega}{\omega_1}}{1+j\frac{\omega}{\omega_1}} \cdot \frac{j\frac{\omega}{\omega_2}}{1+j\frac{\omega}{\omega_2}}$ → C_G i C_D utječu na niskim frekv (za $\omega \ll \omega_3$)

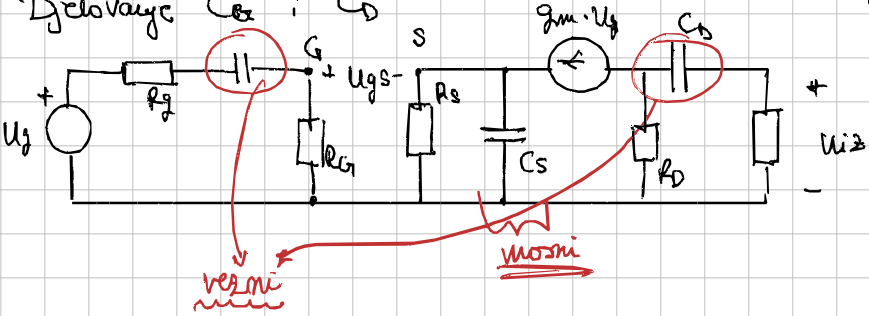
Mosni kond $\frac{1+j\frac{\omega}{\omega_3}}{1+j\frac{\omega}{\omega_4}}$ } kreću se kratko spajati pri „visokoj" frekvenciji → $\tau_3 = C \cdot R_{(pri)}$ pri o kojoj vidi

Pri analizi pretpostavljamo da smo dovoljno visoko da onaj drugi krade K.S. a da ovaj premostrani još djeluje

→ ω_4 (normalno raćuna) → pri kojoj ω se kapacitet skroz kratko spaja

C_S — $\omega_3 \to$ nula; $\omega_4 \to$ pol

Djelovanje C_G i C_D



$\omega \ll \omega_3$ $\left(\frac{1}{0} = \infty\right)$

↳ R_S djeluje

$\omega \gg \omega_4$

↳ R_S ne djeluje (K.S.)

Kada djeluju C_G i C_D djeluju na $\omega \ll \omega_3$ kada C_E još ne djeluje

Svaki od serijskih kapaciteta RUŠI pojačanje na NF.

↳ pojedini kapacitet počinje smanjivati pojačanje u okolišu ω na kojoj je impedancija kapaciteta jednaka otporu priključne vodiče konst.

⇒ Smanjivanjem ω pod pojačanje je stalan i pojačanje teži u $\emptyset$.

C_G i C_D djeluju na $\omega \to 0$!

Izraz za pojačanje: $A_v(j\omega) = A_{v0} \cdot \underbrace{\frac{1+j\frac{\omega}{\omega_1}}{1+j\frac{\omega}{\omega_2}}}_{C_S} \cdot \underbrace{\frac{j\frac{\omega}{\omega_3}}{1+j\frac{\omega}{\omega_3}}}_{C_G} \cdot \underbrace{\frac{j\frac{\omega}{\omega_4}}{1+j\frac{\omega}{\omega_4}}}_{C_D}$

C_D i C_G

ruše pojačanje

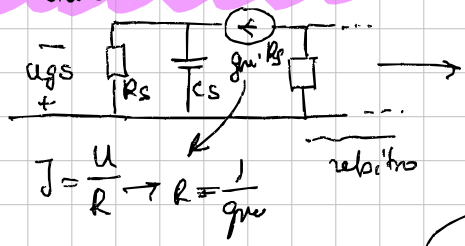
djeluju kada je CE odspojen ---> postoji degeneracija

Doprinos C_S: umjesto da ručno izvodimo A_v gledati što C_S vidi

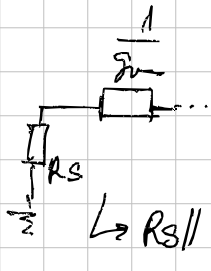
$$\omega_1 = \frac{1}{\tau_{S1}} = \frac{1}{C_S \cdot R_S} \quad \text{(uvijek)}$$

Za sve ostale: K.S. → $u_{ul} = 0$

ω_2:



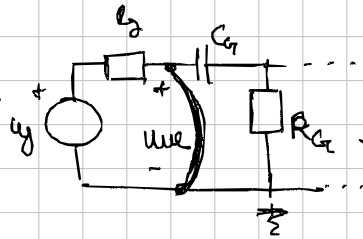
$I = \frac{U}{R} \rightarrow R = \frac{1}{g_m}$

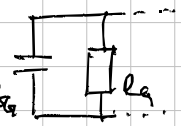


↳ $R_S \parallel \frac{1}{g_m} = R'$

$$\Rightarrow \omega_2 = \frac{1}{\tau_{S2}} = \frac{1}{C_S(R_S \parallel \frac{1}{g_m})}$$

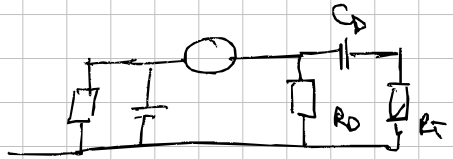
$\omega_3 \Rightarrow$

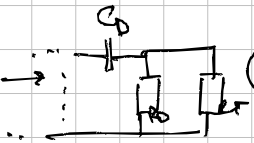




$$\Rightarrow \omega_3 = \frac{1}{\tau_G} = \frac{1}{C_G \cdot R_G}$$

$\omega_4 \Rightarrow$





$$\omega_4 = \frac{1}{C_D(R_D + R_T)}$$

↳ onda to sve uvrstimo u $A_v(j\omega)$